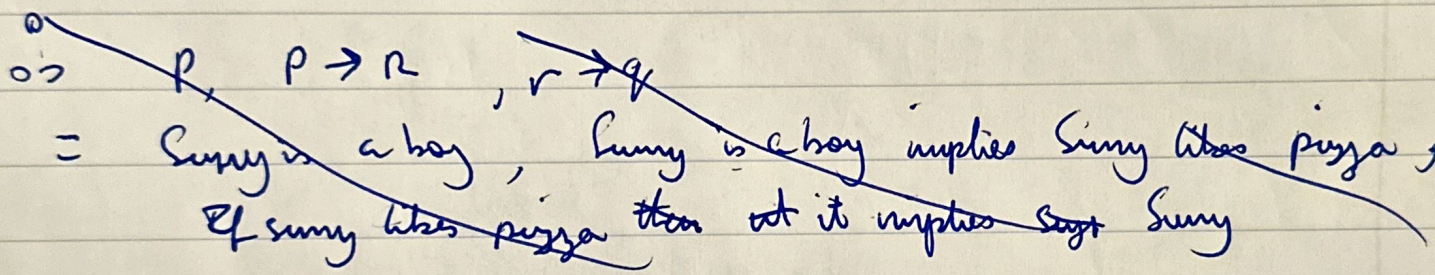
1.) Where p: Sunny is a boy
r: Sunny likes Pizza
q: Sunny likes football



$\therefore$ p, $p \rightarrow r$, $r \rightarrow q$

= Sunny is a boy, Sunny is a boy implies Sunny likes pizza, If sunny likes pizza then it implies Sunny

p - Sunny likes Pizza

$p \rightarrow r$ - Sunny likes pizza implies Sunny likes pizza is a boy

$r \rightarrow q$ - Sunny likes pizza implies Sunny likes football

$\therefore$ Sunny likes pizza $\equiv \therefore q$

$\therefore$ Argument is valid.

2.) $$\frac{P \to (q \to r),\ q \to (P \to r)}{\therefore (P \vee q) \to r}$$

P: Sunny is a boy

Q: Sunny likes ~~pizza~~ likes football

R: Sunny likes pizza

$P \to (q \to R)$

≡ Sunny is a boy implies that Sunny liking ~~pizza~~ pizza implies sunny likes football

$q \to (P \to r)$

≡ Sunny likes football implies that Sunny being a boy implies that sunny likes pizza

$\therefore$ the union of sunny being a boy & Sunny liking football must imply that sunny ~~likes pizza~~ likes pizza

$\therefore (P \vee q) \to r$ is Valid.